

MATH 1210 Worksheet 1 Summer 2023

Suggested exercises, more exercises can be found in the textbook or on Dr. R.S.D. Thomas' webpage for archived material:

<http://home.cc.umanitoba.ca/~thomas/Courses>

Questions marked with * are more difficult.

1. Use induction to show that for $n \geq 1$ we have

$$1^2 + 2^2 + 3^2 + \cdots + n^2 = \frac{n(n+1)(2n+1)}{6}.$$

2. Use induction to show that for $n \geq 1$ we have

$$1^3 + 2^3 + 3^3 + \cdots + n^3 = \frac{n^2(n+1)^2}{4}.$$

3. Use induction to show that for $n \geq 1$ we have

$$1 + 3 + 6 + 10 + \cdots + \frac{n(n+1)}{2} = \frac{n(n+1)(n+2)}{6}.$$

4. Use induction to show that for $n \geq 1$ we have

$$2 + 2^3 + 2^5 + \cdots + 2^{2n-1} = \frac{2(2^{2n} - 1)}{3}.$$

5. * Use induction to show that for $n \geq 1$ we have

$$(3n+1) + (3n+4) + (3n+7) + \cdots + (6n-2) = \frac{n(9n-1)}{2}.$$

6. Use induction to show that 7 divides $8^n - 1$ for $n \geq 1$.

7. * Find the fallacy in the following inductive argument to prove that "all birds are the same colour":

Base case when $n = 1$: *In a set of birds consisting of only one bird, all birds are clearly the same colour. Hence the result is valid when the number of birds in a set is $n = 1$.*

Induction step: *Suppose now that the result is valid when a set of birds contains k birds; that is all k birds are the same colour. Consider any set of $k+1$ birds, and enumerate the birds with labels $1, 2, \dots, k+1$. The set of birds with labels $1, 2, \dots, k$ must all be the same colour by the inductive hypothesis. The set of birds $2, 3, \dots, k+1$ must also all be the same colour. Since these two sets have $k-1$ birds in common (namely the one with labels $2, 3, \dots, k-1$), all $k+1$ birds must have the same colour.*

Conclusion: *Thus, by the principle of mathematical induction, any set of n birds must all be the same colour.*

8. * Use induction to show that for $n \geq 1$ we have

$$1 + 3 + 5 + \cdots + (2n-1) < n^2 + n.$$

9. * Use induction to show that $9(n!) > 2^{2n}$ for $n \geq 5$.

10. Express the sum in sigma notation

$$\frac{1}{2} + \frac{2}{4} + \frac{3}{8} + \frac{4}{16} + \frac{5}{32} + \cdots + \frac{10}{1024}.$$

Initiate the summation with the integer 1.

11. Express the sums that appear in questions 1,2, 3,4,5 in sigma notation.

12. Express the sum in sigma notation

$$2 \cdot 3 + 3 \cdot 4 + 4 \cdot 5 + \cdots + 99 \cdot 100.$$

Initiate the summation with the integer 1.

13. Verify that the two summations are identical

$$\sum_{j=1}^{14} 3^{j+5}(j+4)! \quad \text{and} \quad \sum_{j=5}^{18} 3^{j+1}j!$$

14. Use the formulas of questions 3 and 4 to evaluate the sum

$$\sum_{k=1}^5 \frac{4^k + k(k+1)}{2}.$$

15. Use the formulas of questions 1 and 2 to evaluate the sum

$$\sum_{k=2}^{29} (k^3 - 3k^2).$$

16. * Find a formula for the summation

$$\sum_{k=1}^n \frac{1}{k(k+1)}.$$

Hint: $\frac{1}{k(k+1)} = \frac{1}{k} - \frac{1}{k+1}.$